



Cloud Computing

Assignment-Week 1

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 10

Total mark: 10 X 1 = 10

QUESTION 1:

Which of the following best represents the meaning of resource pooling in cloud computing?

- A. Every customer receives a permanently dedicated physical server.
- B. Computing resources are dynamically assigned to multiple customers using a shared infrastructure.
- C. Resources can only be accessed from a local network.
- D. Users must manually allocate hardware before using cloud services

Correct Answer: B.

Detailed Solution: Resource pooling is one of the NIST essential characteristics of cloud computing. Providers serve multiple customers using shared computing resources that are dynamically assigned and reassigned according to demand.

QUESTION 2:

A startup rents virtual servers on Amazon EC2. The team installs and patches its own operating system, configures the web server, and manages the runtime and applications itself. The provider supplies only the virtualized compute, storage, and networking. This is best classified as:

- A. SaaS
- B. PaaS
- C. IaaS
- D. FaaS

Correct Answer: C

Detailed Solution: The consumer manages the operating system, runtime, and applications while the provider supplies virtualized compute, storage, and networking. This division of responsibility is the defining feature of Infrastructure as a Service (IaaS).

QUESTION 3:

Which deployment model allows an organization to maintain cloud infrastructure exclusively for its own use, either on-premises or hosted by a third party?



- A. Public Cloud
- B. Hybrid Cloud
- C. Community Cloud
- D. Private Cloud

Correct Answer: D

Detailed Solution:

A private cloud is dedicated to a single organization, providing greater control over security, compliance, and resource management.

QUESTION 4:

Virtualization technology allows multiple virtual machines (VMs) to run on a single physical system or on a cluster of physical systems. Which of the following statements is the MOST accurate?

- A. Each VM always shares the same operating system kernel with the host OS.
- B. Multiple VMs can run different operating systems on the same physical machine simultaneously.
- C. Virtualization is only possible in cloud computing environments.
- D. Virtualization prevents resource overcommitment on the host system.

Correct Answer: B

Detailed Solution: Modern hypervisors (e.g., VMware, KVM) allow multiple VMs with different OSs to share the same physical hardware. Option A describes containers more than full VMs. Virtualization pre-dates cloud (so C is false), and resource overcommitment (CPU/memory overcommit) is actually a feature often used, not prevented (D is false).

QUESTION 5:

Consider the following statements about distributed systems (DS):

- (i) DS can improve scalability by adding more nodes to the system.
- (ii) DS can increase fault tolerance through data and service replication.
- (iii) DS guarantees lower latency for every single operation compared to a centralized system.

Choose the MOST APPROPRIATE option:

- A. (i) and (ii)
- B. Only (iii)
- C. (i) and (iii)



D. (i), (ii), and (iii)

Correct Answer: A

Detailed Solution: Distributed systems improve scalability by adding nodes and increase fault tolerance through replication. However, distribution introduces network communication and coordination overhead, so lower latency for every operation is not guaranteed and some operations may be slower (iii is false).

QUESTION 6:

The combination of Service-Oriented Architecture (SOA) and Cloud Computing led to a paradigm where almost any IT capability can be delivered as a network-accessible service. This is often referred to as:

- A. FTP
- B. SNTP
- C. XaaS
- D. FaaS

Correct Answer: C

Detailed Solution: XaaS = “Anything/Everything as a Service”, an umbrella term for delivering diverse IT services via the cloud. FaaS (Functions as a Service) is only one specific service model, not the broad paradigm.

QUESTION 7:

Which of the following correctly describes a Type-2 (hosted) hypervisor?

- A. It runs directly on the physical hardware without any underlying operating system.
- B. It runs as an application on top of a host operating system.
- C. It always provides lower virtualization overhead than a Type-1 hypervisor.
- D. It cannot run more than one virtual machine at a time.

Correct Answer: B

Detailed Solution: A Type-2 (hosted) hypervisor runs as an application on top of a host operating system (e.g., VirtualBox, VMware Workstation). A Type-1 (bare-metal) hypervisor runs directly on hardware and typically gives lower overhead, so A and C describe Type-1; D is false since hosted hypervisors can run multiple VMs.



QUESTION 8:

Which of the following are advantages of Platform-as-a-Service (PaaS)?

- (i) Developers can focus primarily on application development.
- (ii) Automatic software updates for the underlying platform.
- (iii) Complete control over physical server hardware.
- (iv) Faster application deployment.

- A. (i), (ii), and (iv)
- B. (ii) and (iii)
- C. (i) and (iii)
- D. (i), (ii), (iii), and (iv)

Correct Answer: A

Detailed Solution : PaaS simplifies software development by abstracting infrastructure management and providing managed runtime environments. Developers typically do not control physical hardware.

QUESTION 9:

Which of the following services is most likely to be classified as Infrastructure as a Service (IaaS)?

- A. Microsoft Word Online
- B. Amazon EC2 Virtual Machines
- C. Google Docs
- D. Salesforce CRM

Correct Answer: B

Detailed Solution: Amazon EC2 provides virtual machines where users manage the operating system, software, and applications, making it an IaaS offering.

QUESTION 10:

A _____ is a form of distributed computing in which a large number of geographically dispersed and often heterogeneous resources across multiple administrative domains are loosely coupled and coordinated to work together on a common, large-scale task.

- A. Cluster
- B. Grid



NPTEL Online Certification Courses
Indian Institute of Technology Kharagpur



C. Node

D. Hypervisor

Correct Answer: B

Detailed Solution: Grid computing coordinates geographically dispersed, heterogeneous, and loosely coupled resources spanning multiple administrative domains to solve large-scale tasks. Clusters are tightly coupled and co-located, a node is a single machine, and a hypervisor is virtualization software.